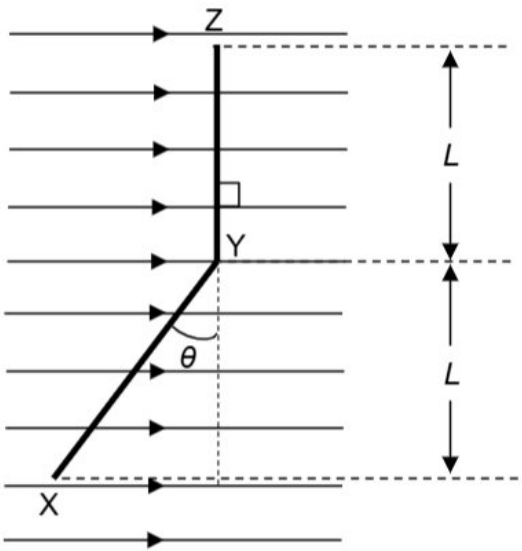
27 A rigid conductor consisting of two straight segments, XY and YZ is placed in a uniform horizontal magnetic field of flux density B as shown. A steady current I flows through the conductor from X to Y to Z.

Segment YZ has length L and is perpendicular to the magnetic field. Segment XY makes an angle θ with the vertical.



The total magnetic force experienced by the entire conductor XYZ is F.

What is the correct expression for the length of segment XY?

A $\dfrac{F}{BI\cos\theta}$ **B** $\dfrac{F}{2BI\cos\theta}$ **C** $\dfrac{F}{BI\sin\theta}$ **D** $\dfrac{F}{2BI\sin\theta}$